Predict Extreme Heat Before It Breaks the Grid

Heat-Pulse

by sPySci

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Problem

Azerbaijan's renewable energy capacity is growing rapidly — from ~5% in 2020 to nearly 10–14% in 2024, with major projects like Garadagh Solar Power Plant and Khizi-Absheron Wind Power Plant expanding grid dependency on weather-sensitive infrastructure. Extreme heat events increase equipment stress, reduce generation efficiency, and raise the risk of transformer failures, grid overloads, and costly outages. Without early forecasting, energy operators remain reactive instead of preventive.

Solution

Heat-Pulse is a 30-day heat risk forecasting system built for Azerbaijan's energy sector.

Using 6 years of historical weather data, we developed a custom Impact Score that combines temperature peaks, heat stress, cooling demand, temperature anomalies, heat persistence, and wind conditions to detect Extreme Heat Events.

This enables Azerenerji and industrial facilities to act early, prevent generator burnouts, and optimize grid load before failures happen.

$$S_{\text{impact}} = 100 \times [0.30(\frac{T_{max}}{T_{ref}}) + 0.20(\frac{HSI}{HSI_{max}}) + 0.20(\frac{CDD}{CDD_{max}}) + 0.15(\frac{P}{3})^2 + 0.10(\frac{|\Delta T|}{A_{max}}) + 0.05(1 - \frac{W_{max}}{W_{ref}})]$$

Data & Pipeline

Multi-Source Data Collection → Pattern & Trend Recognition → 30-Day High-Precision Forecasting → Safety & Impact Scoring → Actionable Strategic Insights

Data Source

Open-Meteo API (Daily & Hourly weather records). Historical and forecast data including air temperature, humidity, wind speed, and solar radiation.

Data Cleaning

Removed 15+ redundant parameters (e.g., soil temperatures, UV index, visibility, and wind/temp measurements at specific altitudes like 80m, 120m, and 180m) and outliers.

Feature Engineering

Transforming raw data into actionable insights like Impact score, Derived Heat Stress Index, Cooling Degree Days (CDD), and Temperature Anomalies to capture localized climate deviations.

ML Model

Model Architecture

We use XGBoost, a gradient boosting ensemble of decision trees that builds models sequentially, where each tree corrects previous errors. It processes structured weather data and captures temporal patterns for accurate heat event prediction.

Why it works?

XGBoost handles nonlinear relationships, resists overfitting through regularization, and performs well on structured datasets. Its efficiency and high accuracy make it ideal for reliable 30-day Extreme Heat Event forecasting.

Forecast

To ensure reliable Extreme Heat Event prediction, we trained multiple forecasting models for key weather and energy variables. Our models showed strong performance across all major indicators.

Temperature forecasting:
achieved R^2 scores between 0.985–0.996, allowing highly accurate heat predictions

Solar radiation forecasting:
reached $R^2 = 0.983$, improving energy demand estimation

Wind speed forecasting:
achieved $R^2 = 0.883$, despite higher environmental variability

Weather classification model:
delivered an F1-score of 0.983, ensuring accurate weather categorization

These outputs feed directly into our Impact Score system, enabling Heat-Pulse to generate a 30-day early warning forecast for Extreme Heat Events. This allows energy operators to prepare in advance, optimize grid load, and reduce the risk of generator failures.

Demo

github repository link

Heat Pulse is an advanced thermal analytics platform designed to bridge the gap between meteorological forecasting and renewable energy management across Azerbaijan's all regions. By integrating real-time weather data with sophisticated statistical models like RMSE and R2, the system provides high-precision insights into solar and wind energy yields while visualizing data through interactive distribution charts and predictive dashboards. Developed with a professional, dark-themed interface, the application empowers energy specialists and data analysts to evaluate regional impact scores and optimize green energy production through science-backed, actionable intelligence.

Impact

Prevent Grid Failures

Early risk detection stops outages before they happen.

Proactive Planning

Enable utilities to schedule load management and maintenance ahead.

Reduce Downtime

Lower operational costs and minimize service interruptions.

Economic Savings

Protect infrastructure investments and reduce emergency response costs.

Thank you!